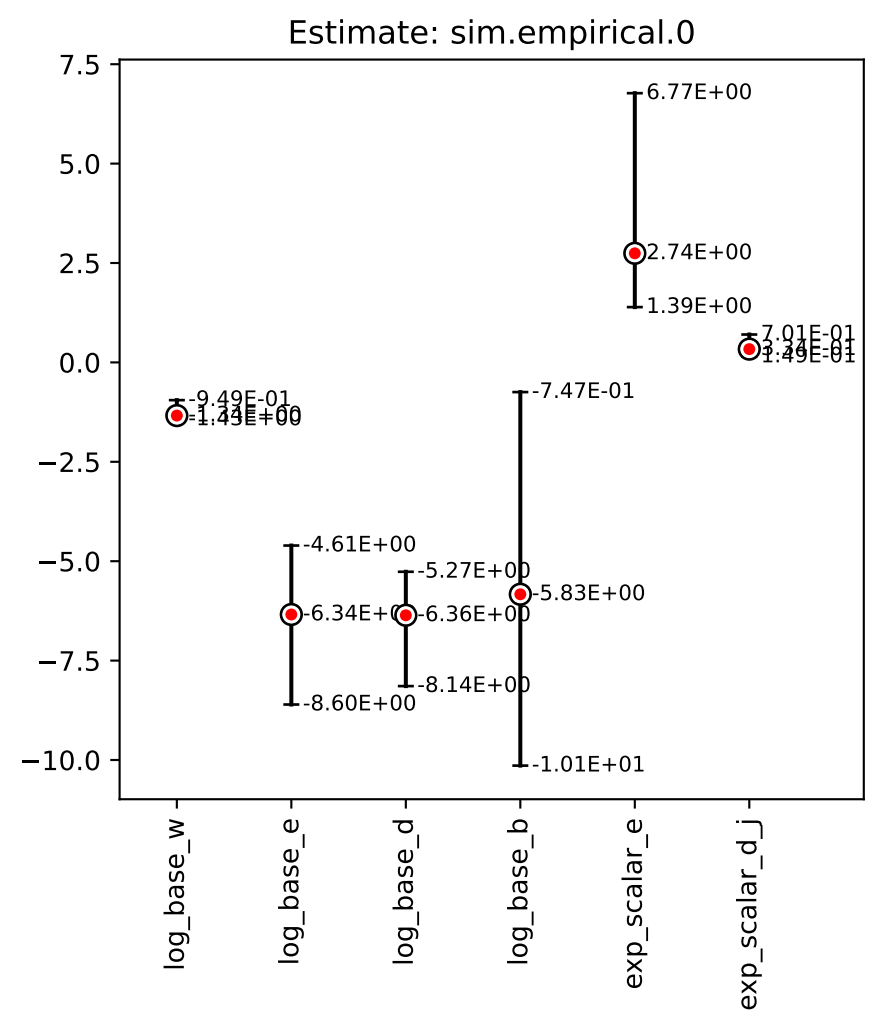
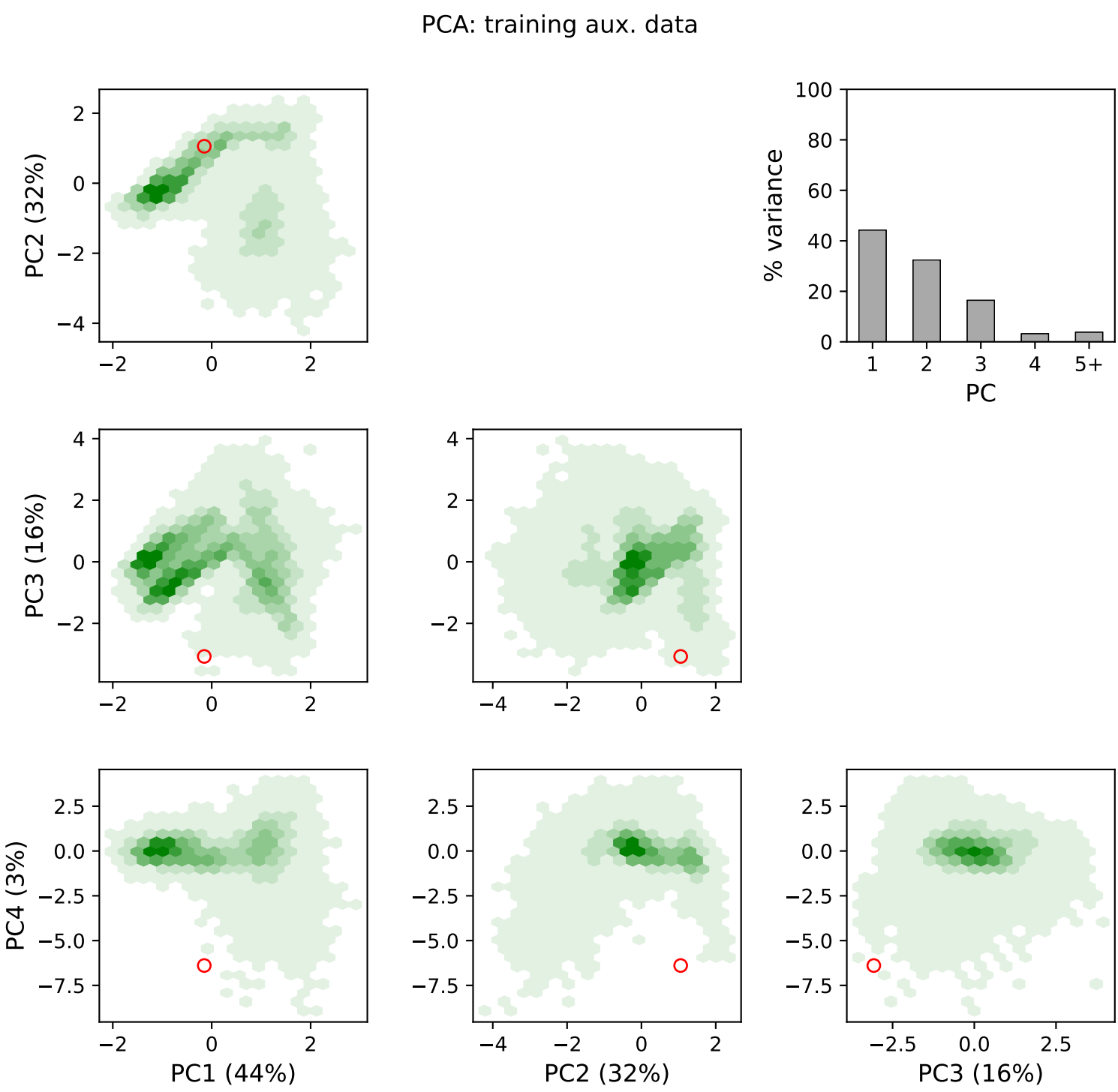
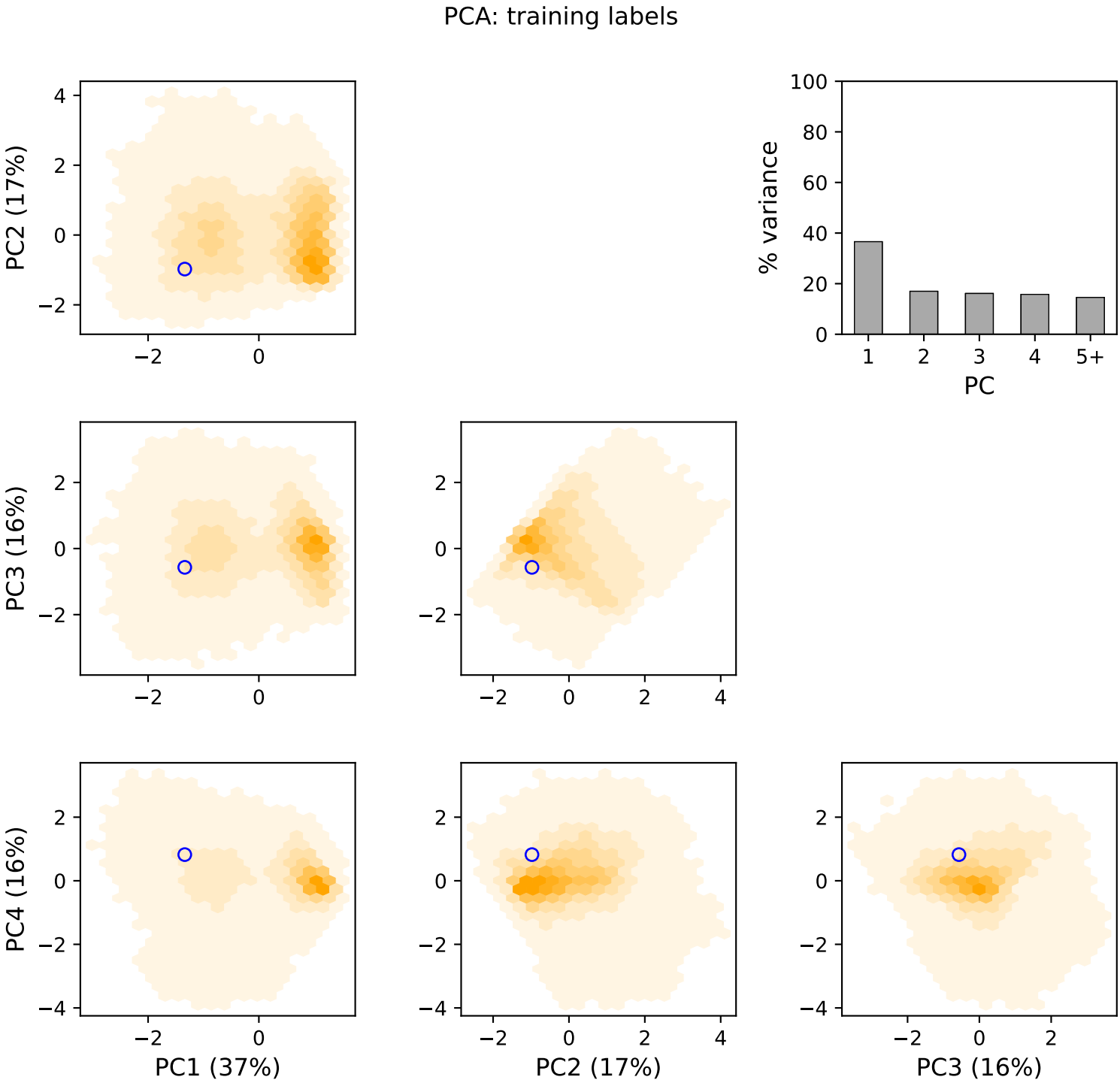
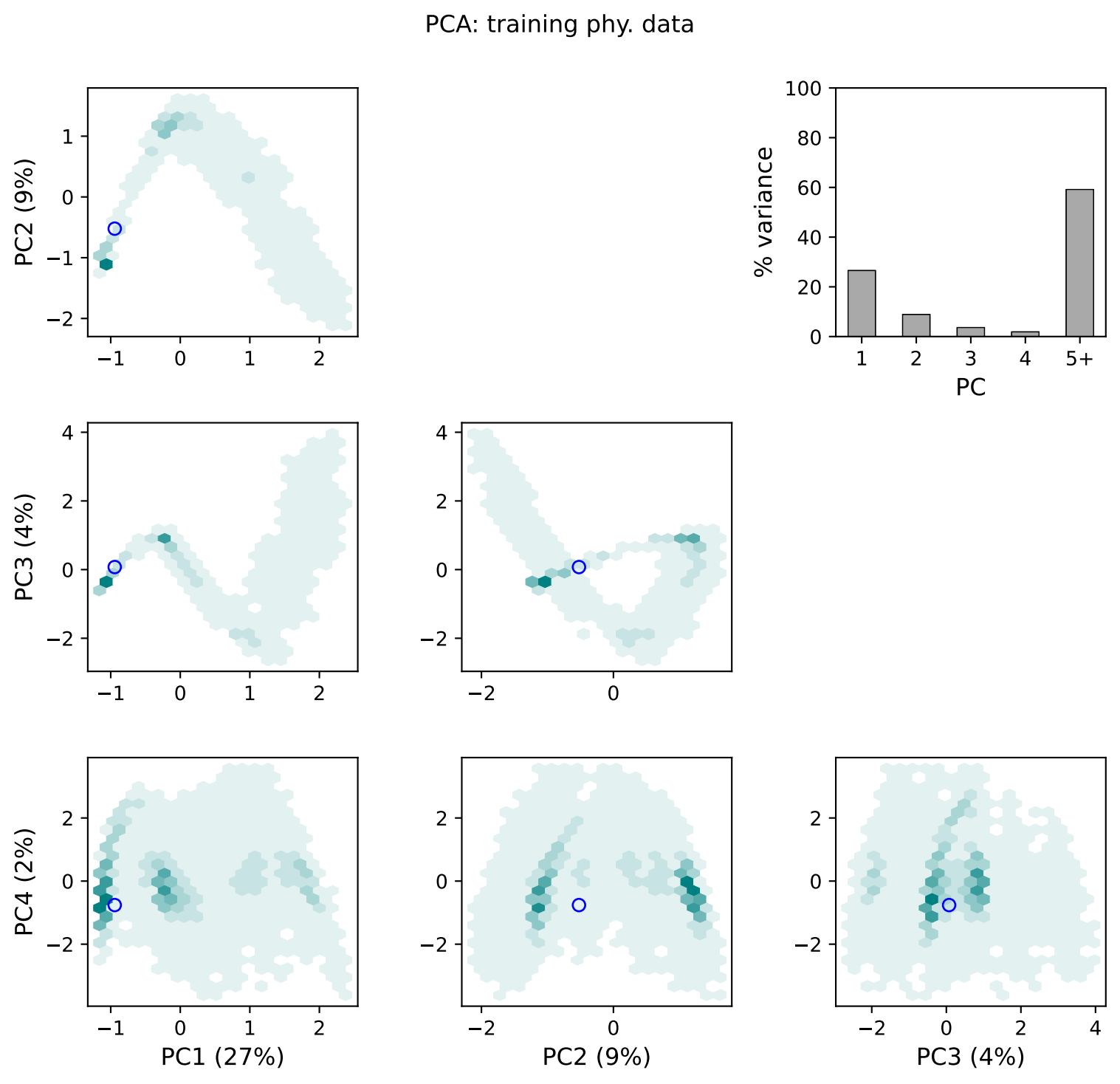
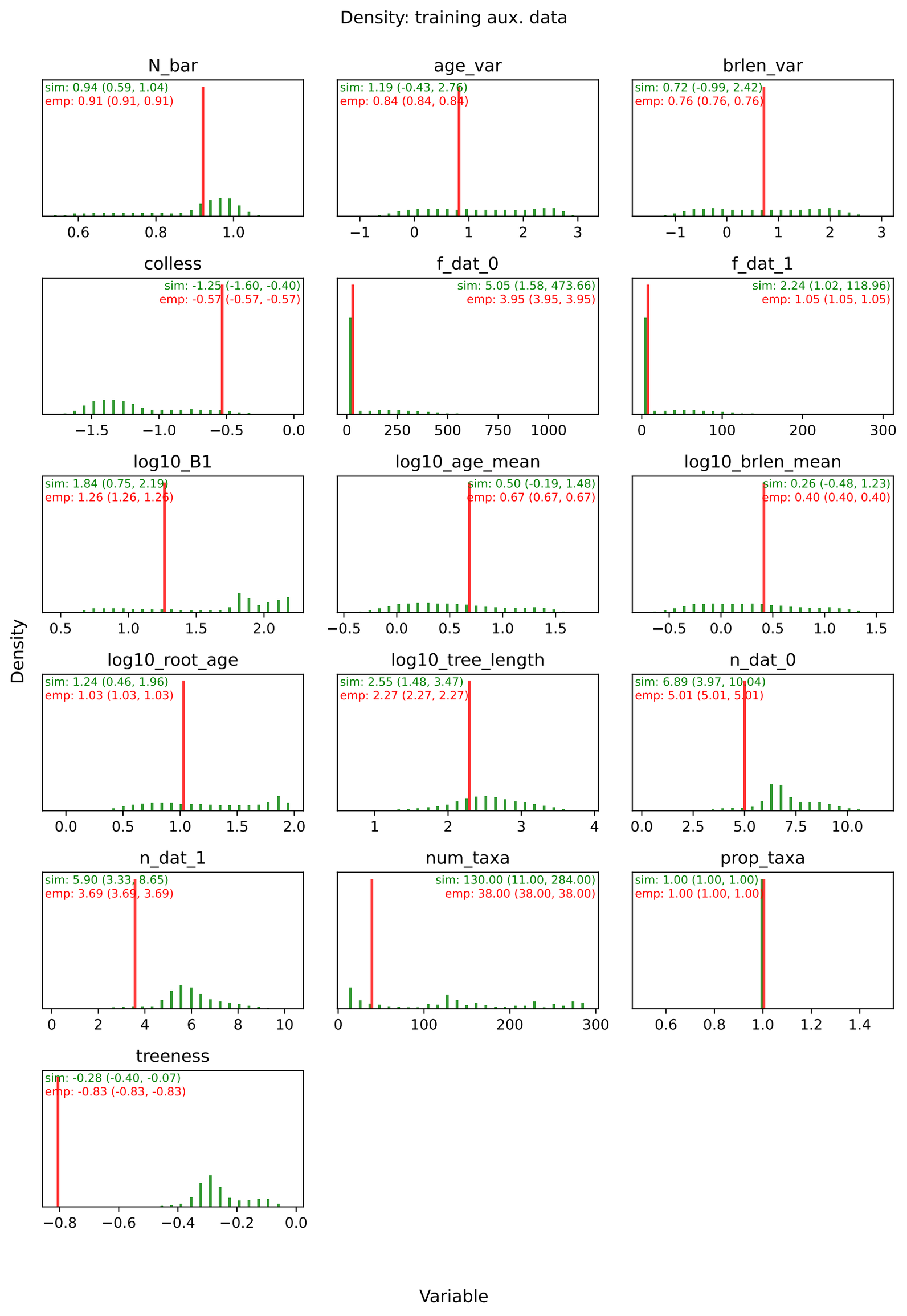
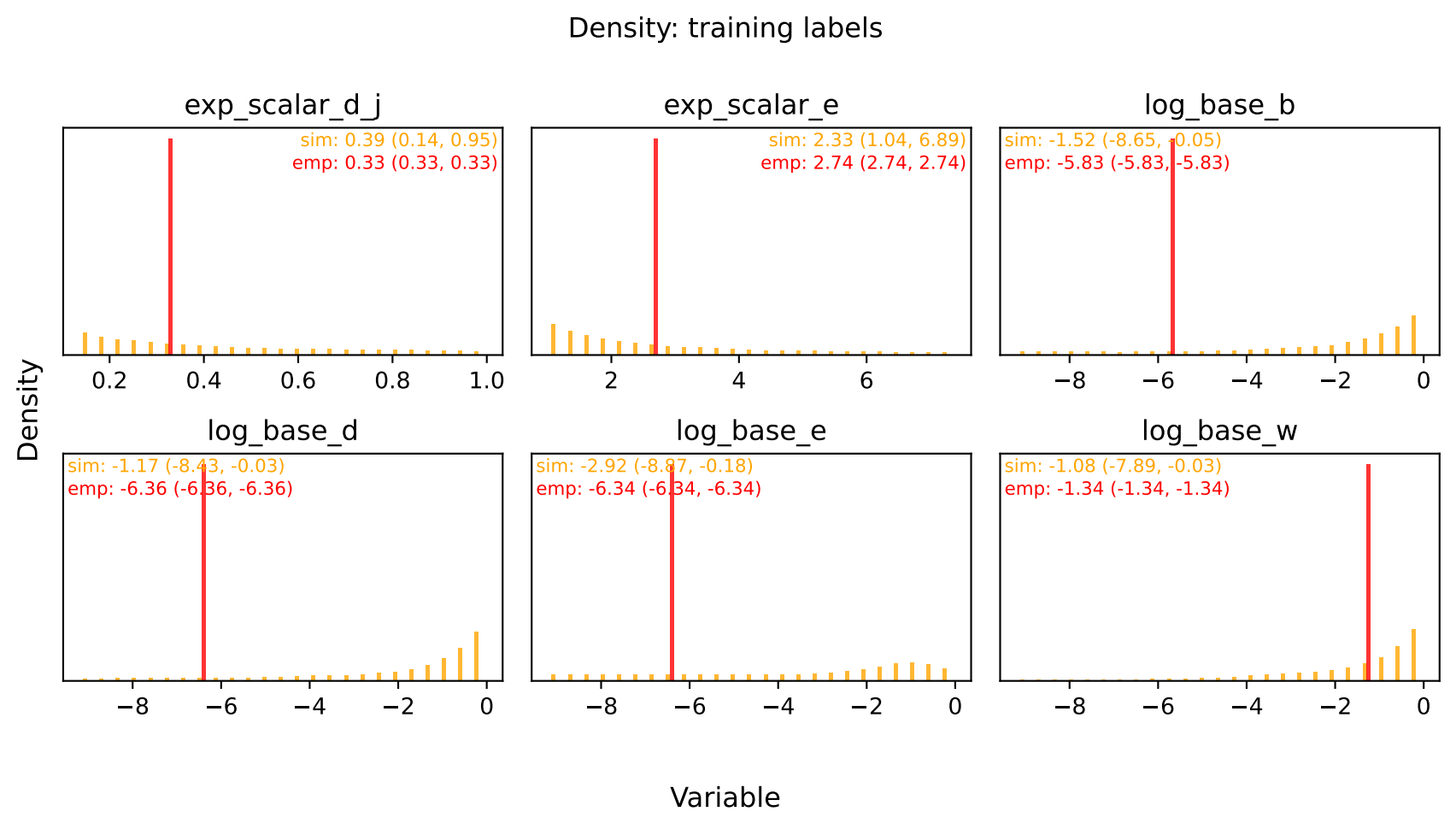

Estimate: sim.empirical.0
7.5
5.0
2.5
0.0
−2.5
−5.0
−7.5
−10.0
6.77E+00
2.74E+00
1.39E+00
7.01E-01
-9.49E-01
-4.61E+00
-5.27E+00
-6.34E+00
-6.36E+00
-5.83E+00
-8.14E+00
-8.60E+00
-7.47E-01
-1.01E+01
log_base_w
log_base_e
log_base_d
log_base_b
exp_scalar_e
exp_scalar_d_j

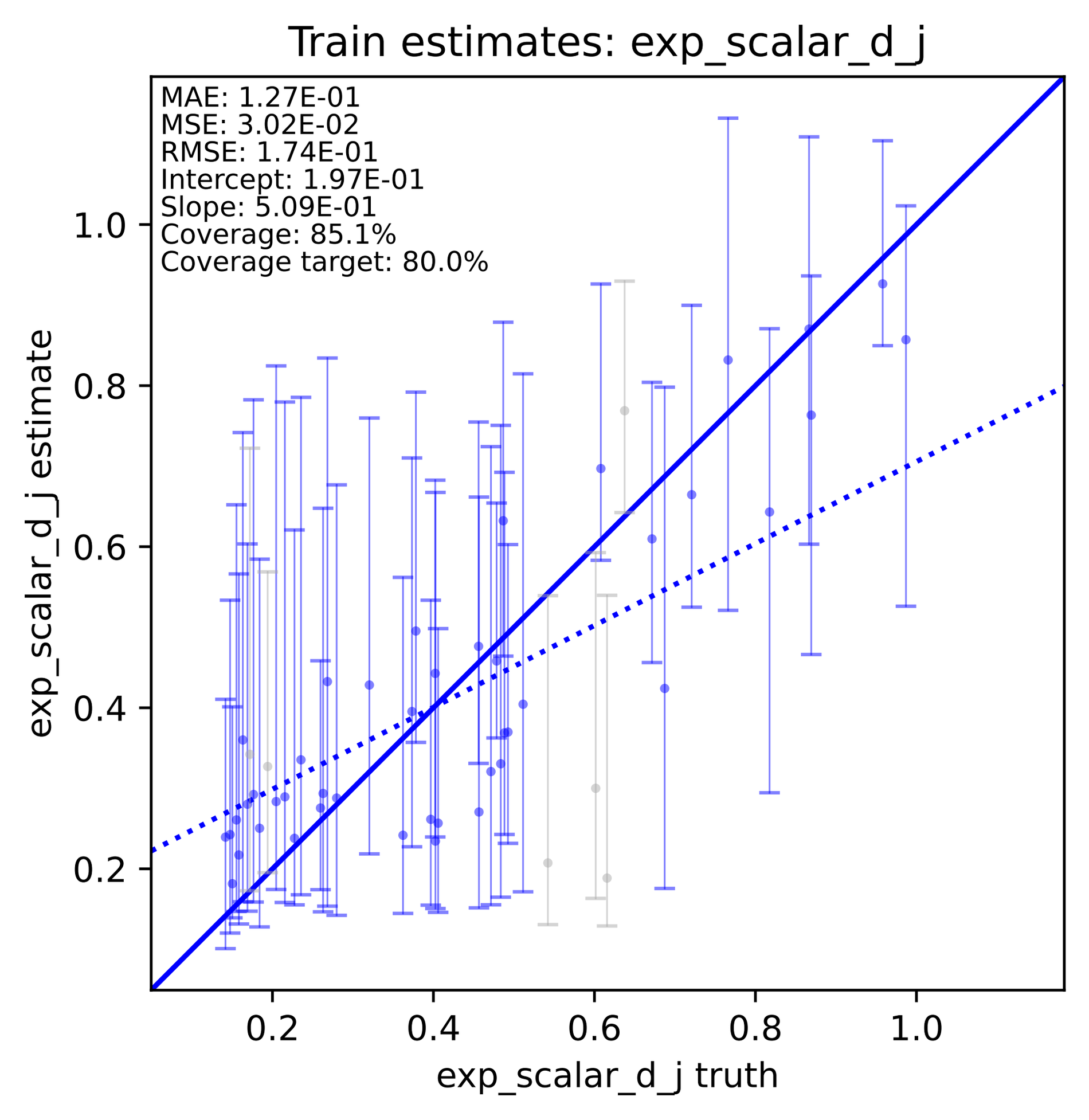

PCA: training aux. data
PC2 (32%)
% variance
PC
PC3 (16%)
PC4 (3%)
PC1 (44%)
PC2 (32%)
PC3 (16%)

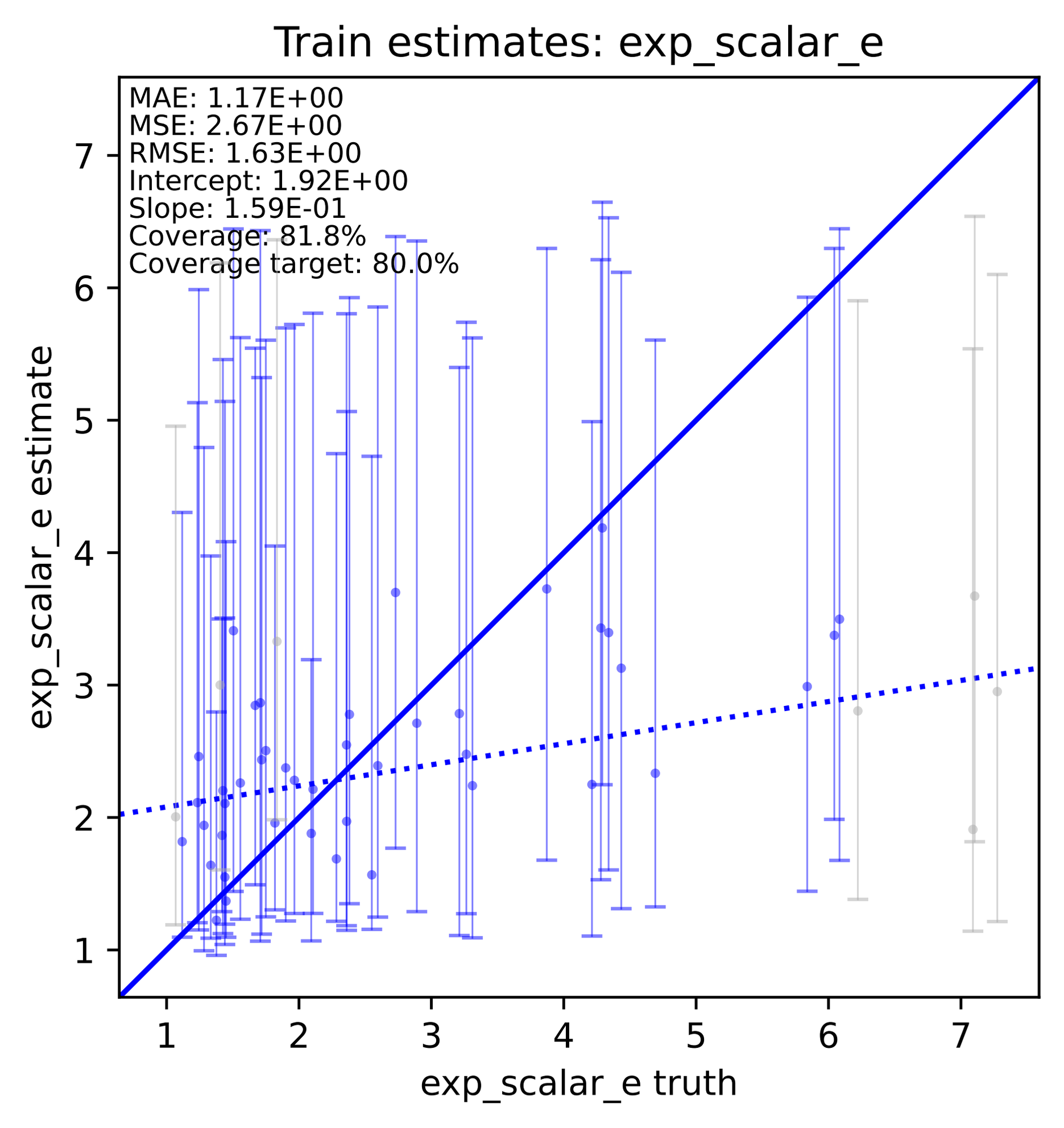

PCA: training labels
PC2 (17%)
% variance
PC
PC3 (16%)
PC4 (16%)
PC1 (37%)
PC2 (17%)
PC3 (16%)

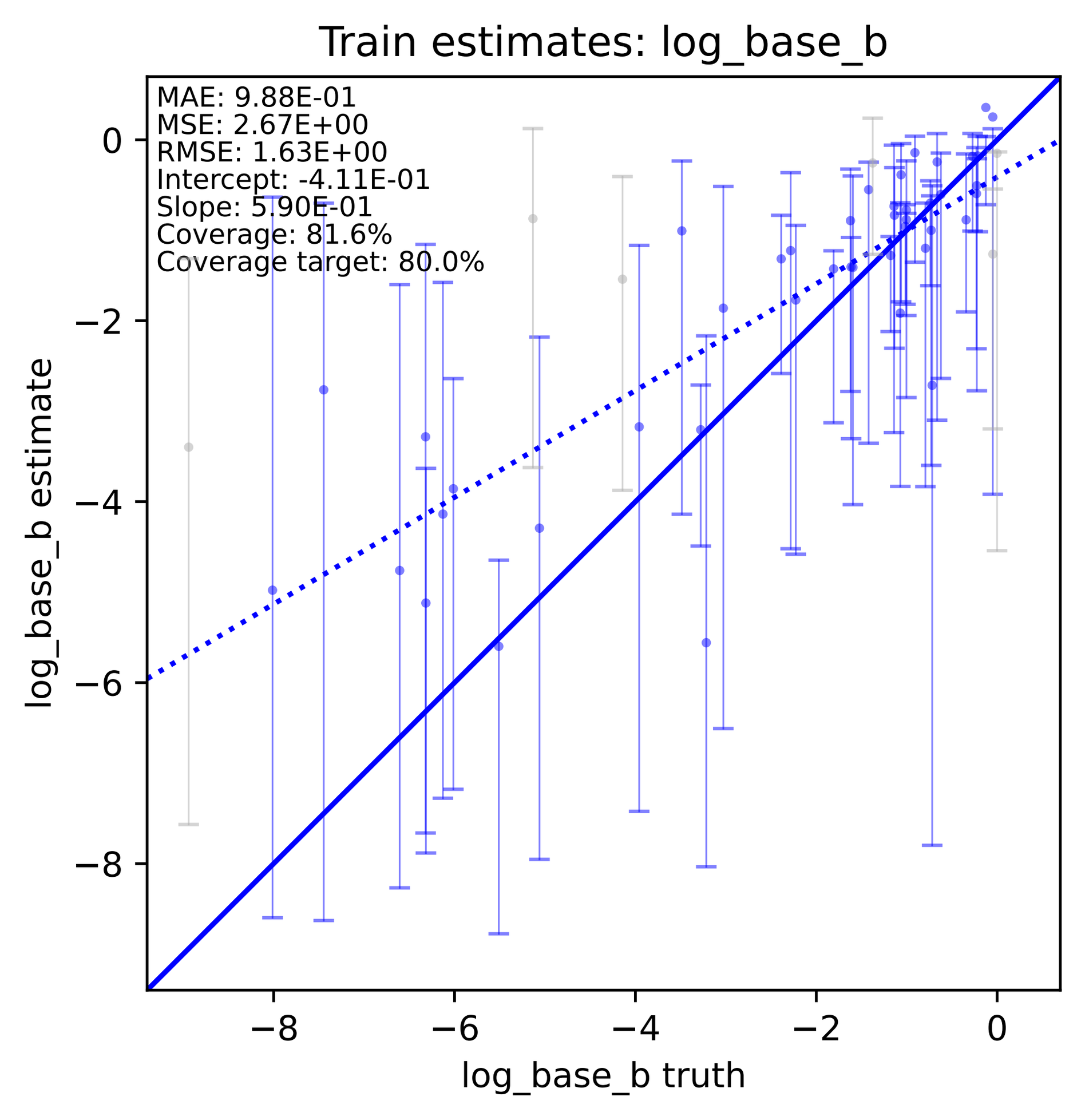

PCA: training phy. data
PC2 (9%)
% variance
PC
PC3 (4%)
PC4 (2%)
PC1 (27%)
PC2 (9%)
PC3 (4%)

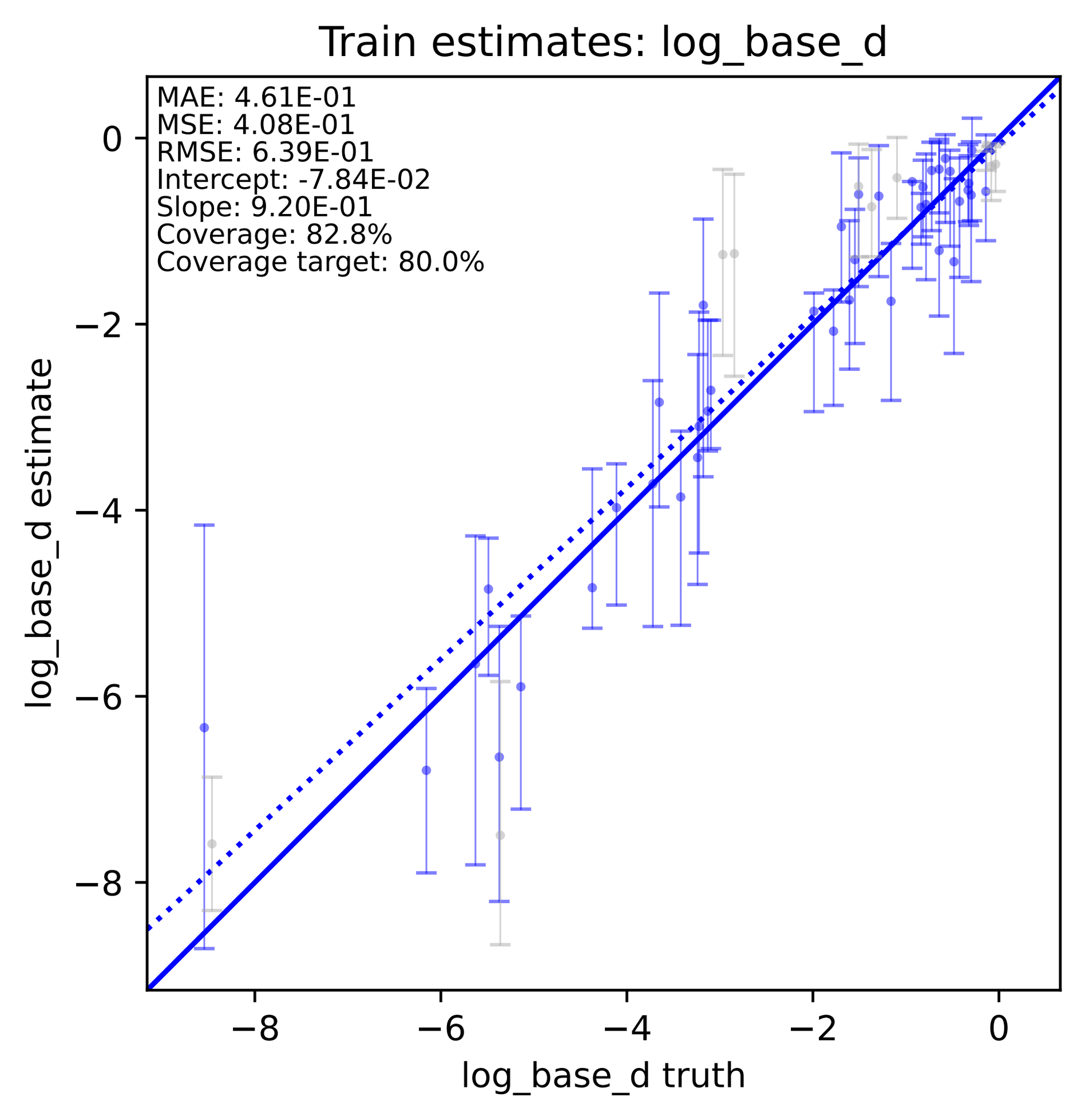

Density: training aux. data
N_bar
sim: 0.94 (0.59, 1.04)
emp: 0.91 (0.91, 0.91)
age_var
sim: 1.19 (-0.43, 2.76)
emp: 0.84 (0.84, 0.84)
brlen_var
sim: 0.72 (-0.99, 2.42)
emp: 0.76 (0.76, 0.76)
colless
sim: -1.25 (-1.60, -0.40)
emp: -0.57 (-0.57, -0.57)
f_dat_0
sim: 5.05 (1.58, 473.66)
emp: 3.95 (3.95, 3.95)
f_dat_1
sim: 2.24 (1.02, 118.96)
emp: 1.05 (1.05, 1.05)
log10_B1
sim: 1.84 (0.75, 2.19)
emp: 1.26 (1.26, 1.26)
log10_age_mean
sim: 0.50 (-0.19, 1.48)
emp: 0.67 (0.67, 0.67)
log10_brlen_mean
sim: 0.26 (-0.48, 1.23)
emp: 0.40 (0.40, 0.40)
log10_root_age
sim: 1.24 (0.46, 1.96)
emp: 1.03 (1.03, 1.03)
log10_tree_length
sim: 2.55 (1.48, 3.47)
emp: 2.27 (2.27, 2.27)
n_dat_0
sim: 6.89 (3.97, 10.04)
emp: 5.01 (5.01, 5.01)
n_dat_1
sim: 5.90 (3.33, 8.65)
emp: 3.69 (3.69, 3.69)
num_taxa
sim: 130.00 (11.00, 284.00)
emp: 38.00 (38.00, 38.00)
prop_taxa
sim: 1.00 (1.00, 1.00)
emp: 1.00 (1.00, 1.00)
treeness
sim: -0.28 (-0.40, -0.07)
emp: -0.83 (-0.83, -0.83)
Density
Variable

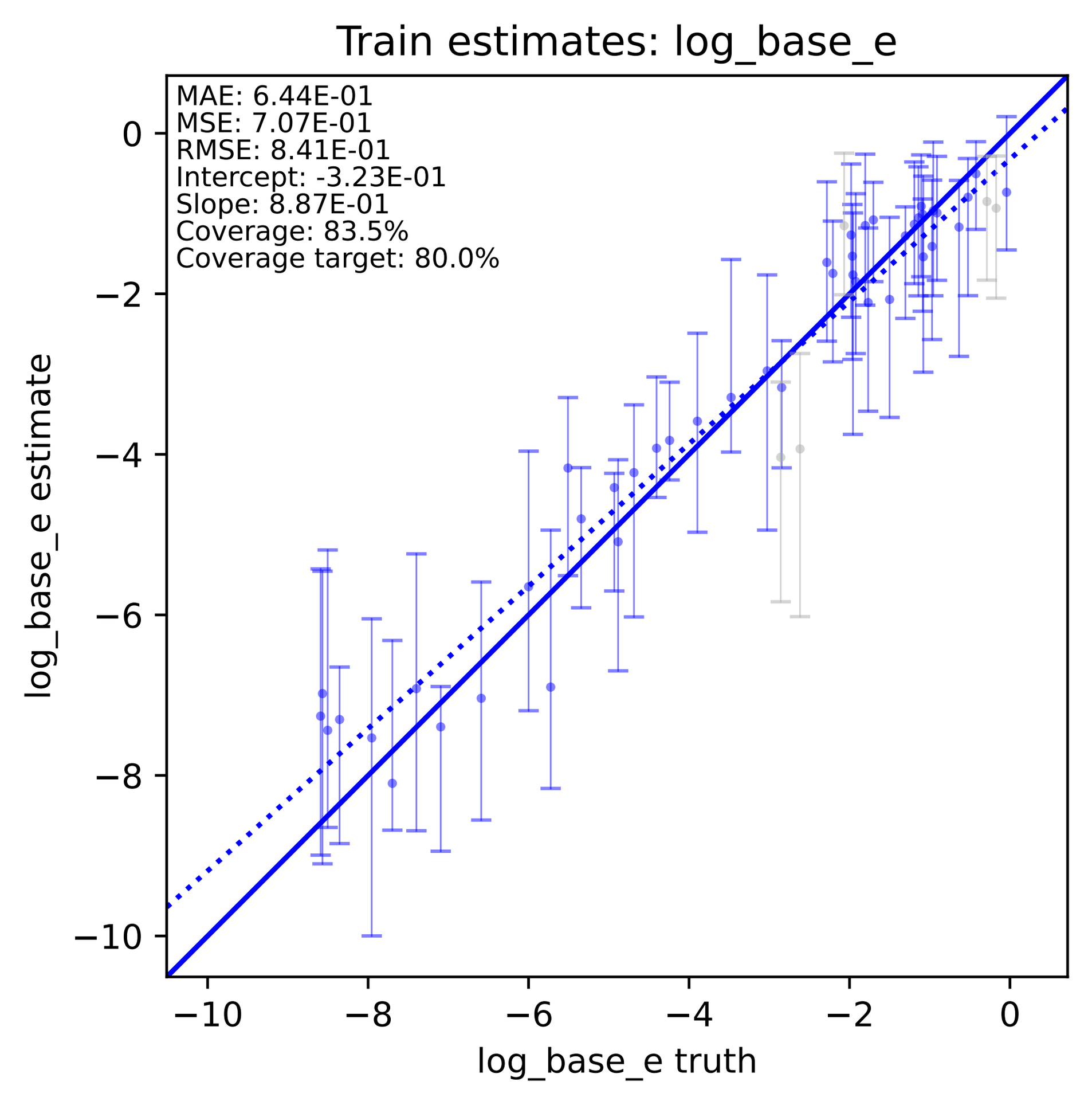

Density: training labels
exp_scalar_d_j
sim: 0.39 (0.14, 0.95)
emp: 0.33 (0.33, 0.33)
exp_scalar_e
sim: 2.33 (1.04, 6.89)
emp: 2.74 (2.74, 2.74)
log_base_b
sim: -1.52 (-8.65, -0.05)
emp: -5.83 (-5.83, -5.83)
log_base_d
sim: -1.17 (-8.43, -0.03)
emp: -6.36 (-6.36, -6.36)
log_base_e
sim: -2.92 (-8.87, -0.18)
emp: -6.34 (-6.34, -6.34)
log_base_w
sim: -1.08 (-7.89, -0.03)
emp: -1.34 (-1.34, -1.34)
Density
Variable

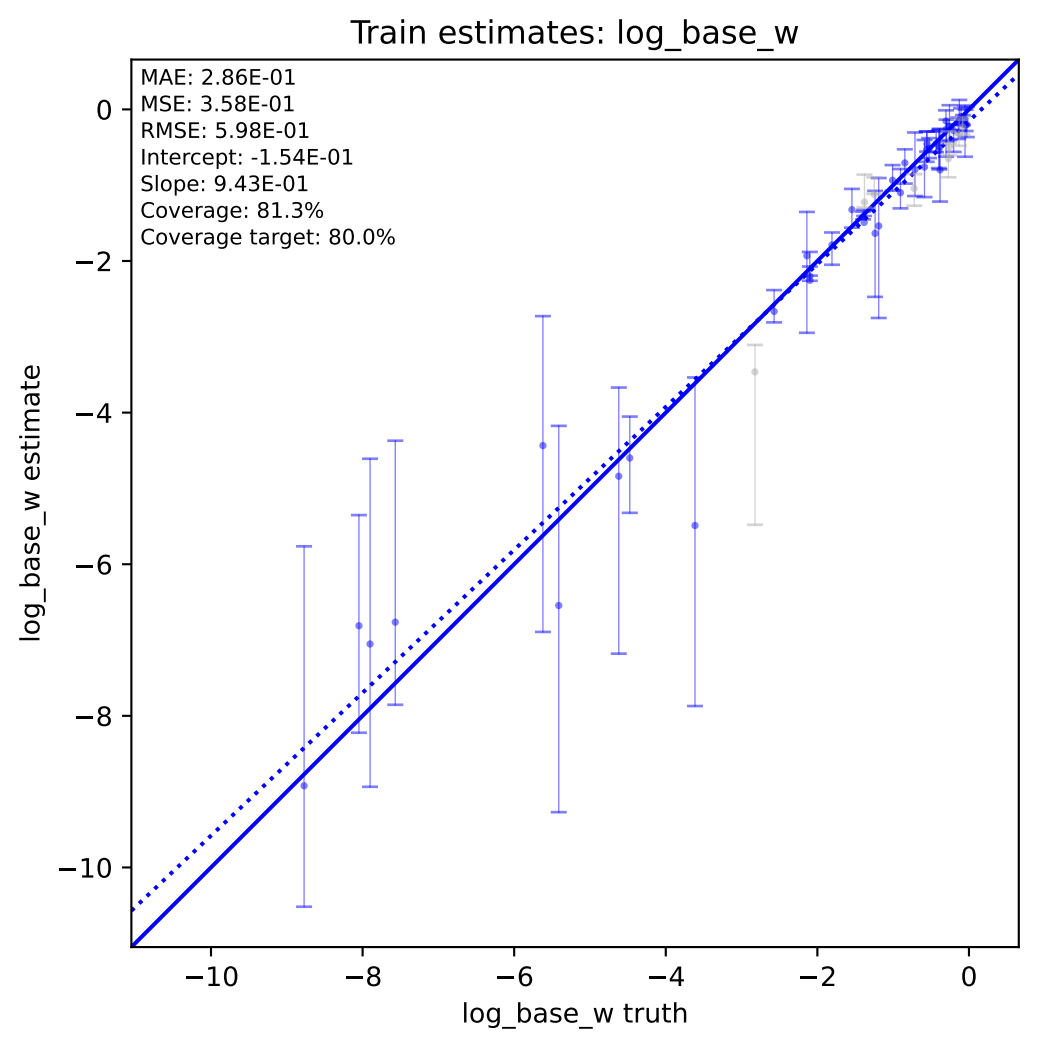

Train estimates: exp_scalar_d_j
MAE: 1.27E-01
MSE: 3.02E-02
RMSE: 1.74E-01
Intercept: 1.97E-01
Slope: 5.09E-01
Coverage: 85.1%
Coverage target: 80.0%
exp_scalar_d_j estimate
exp_scalar_d_j truth
0.2
0.4
0.6
0.8
1.0

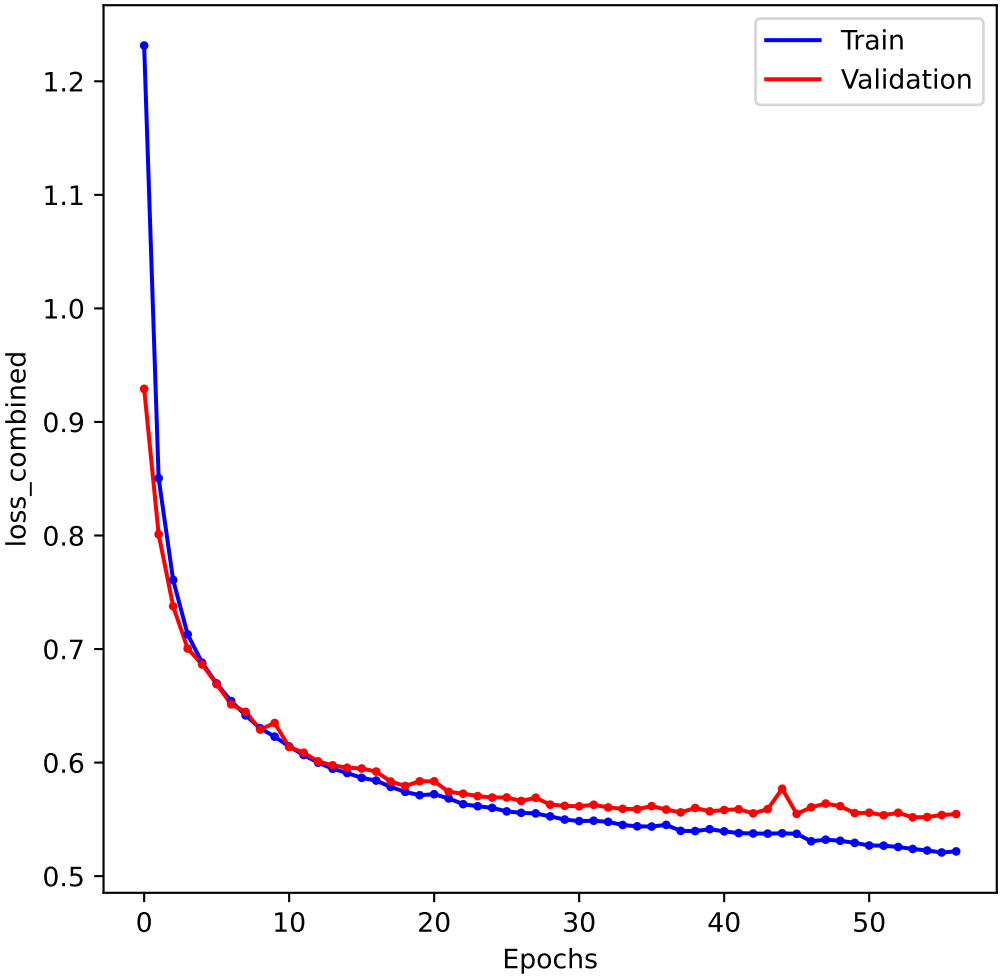

Train estimates: exp_scalar_e
MAE: 1.17E+00
MSE: 2.67E+00
RMSE: 1.63E+00
Intercept: 1.92E+00
Slope: 1.59E-01
Coverage: 81.8%
Coverage target: 80.0%
exp_scalar_e estimate
exp_scalar_e truth
1
2
3
4
5
6
7

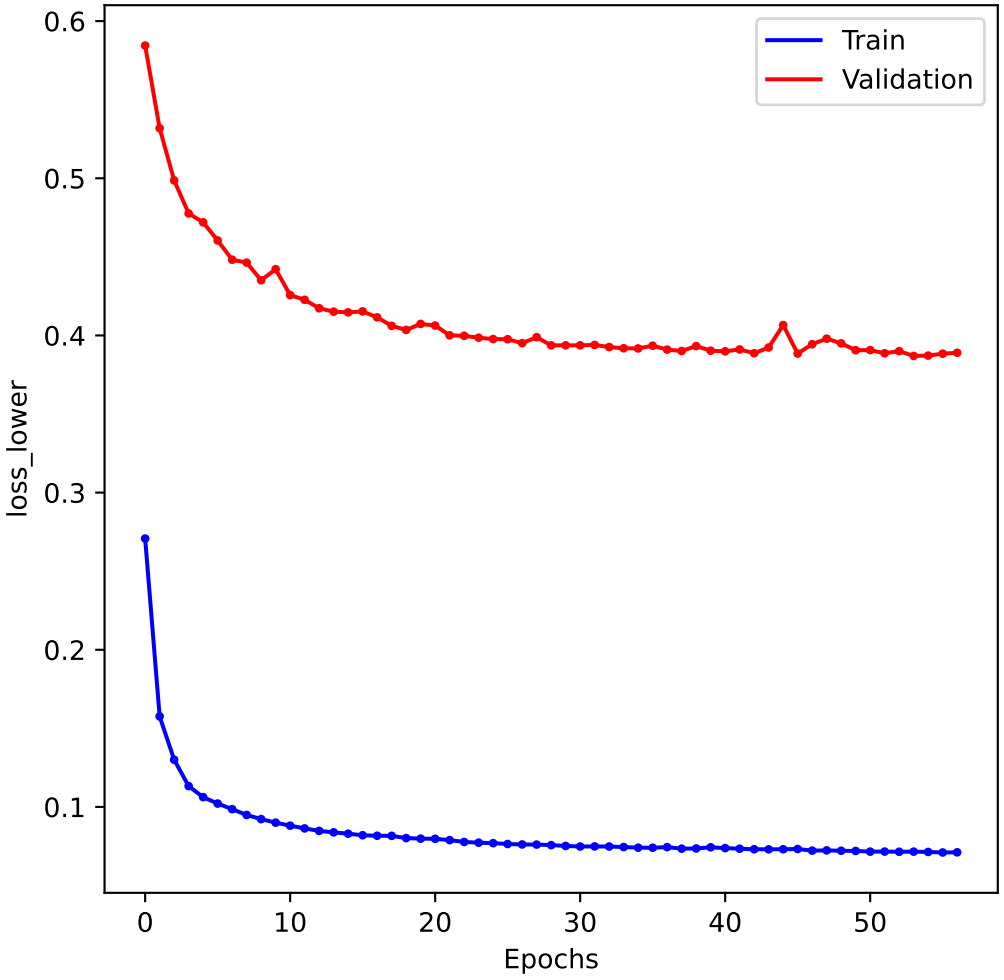

Train estimates: log_base_b
MAE: 9.88E-01
MSE: 2.67E+00
RMSE: 1.63E+00
Intercept: -4.11E-01
Slope: 5.90E-01
Coverage: 81.6%
Coverage target: 80.0%
log_base_b estimate
log_base_b truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

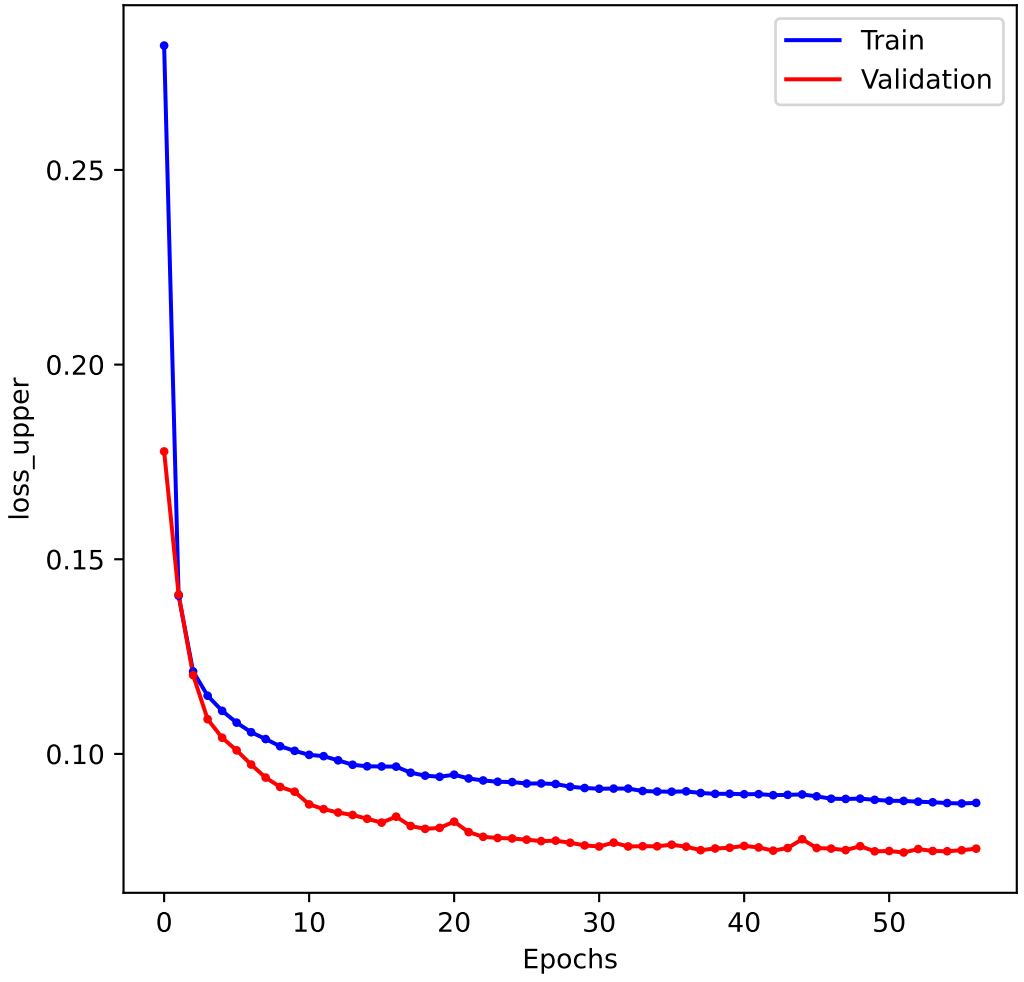

Train estimates: log_base_d
MAE: 4.61E-01
MSE: 4.08E-01
RMSE: 6.39E-01
Intercept: -7.84E-02
Slope: 9.20E-01
Coverage: 82.8%
Coverage target: 80.0%
log_base_d estimate
log_base_d truth
0
−2
−4
−6
−8
−8
−6
−4
−2
0

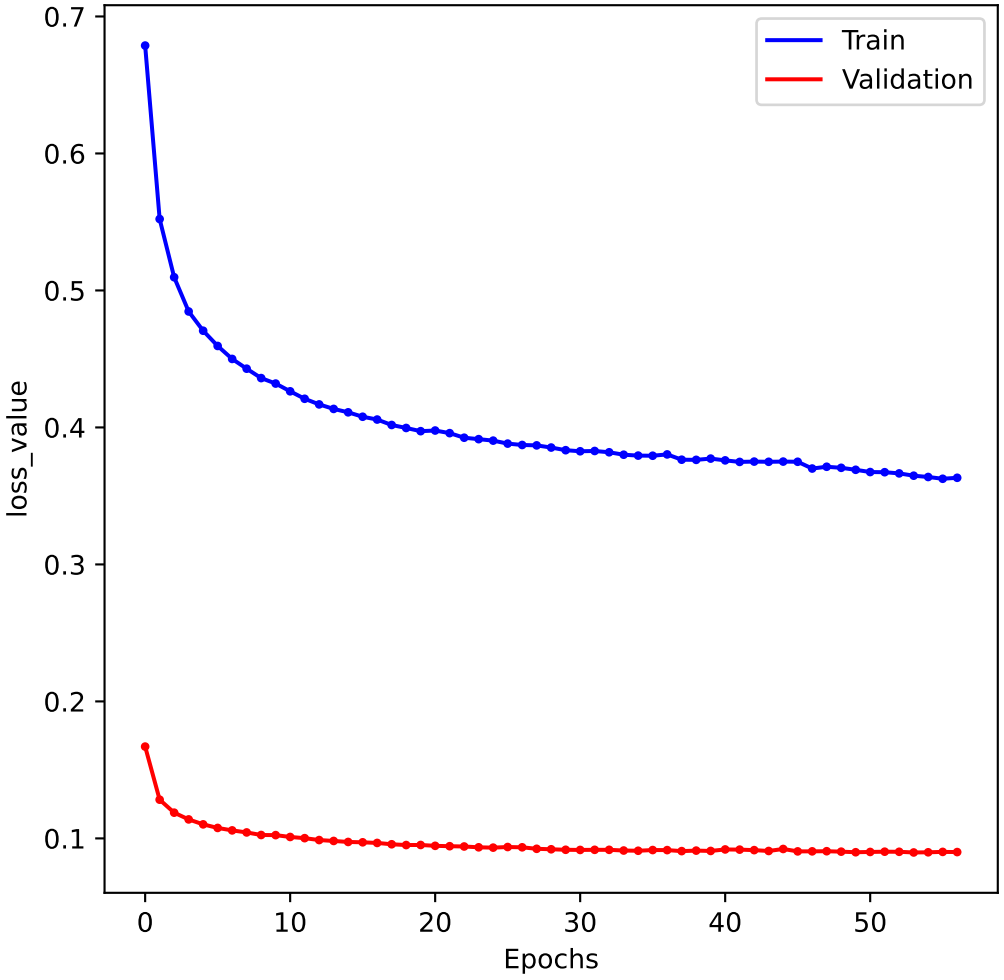

Train estimates: log_base_e
MAE: 6.44E-01
MSE: 7.07E-01
RMSE: 8.41E-01
Intercept: -3.23E-01
Slope: 8.87E-01
Coverage: 83.5%
Coverage target: 80.0%
log_base_e estimate
log_base_e truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

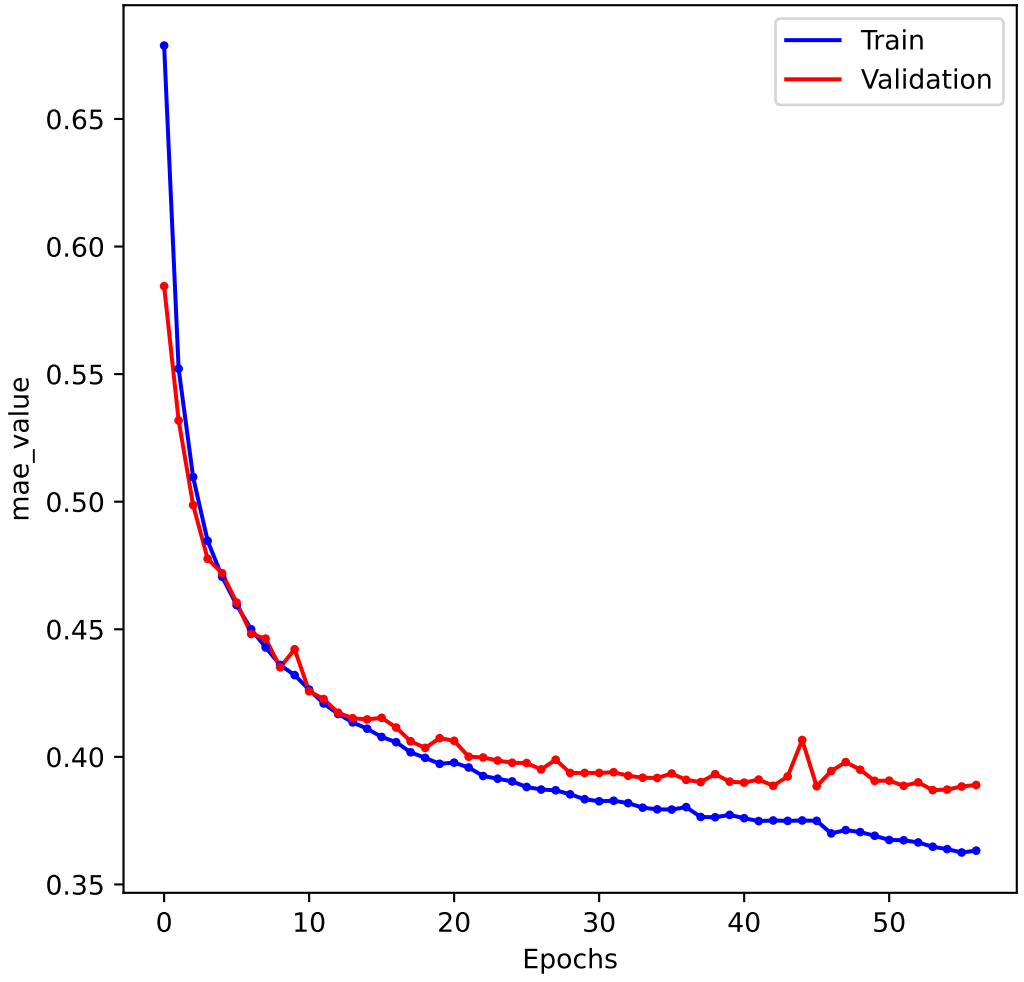

Train estimates: log_base_w
MAE: 2.86E-01
MSE: 3.58E-01
RMSE: 5.98E-01
Intercept: -1.54E-01
Slope: 9.43E-01
Coverage: 81.3%
Coverage target: 80.0%
log_base_w estimate
log_base_w truth
0
−2
−4
−6
−8
−10
−10
−8
−6
−4
−2
0

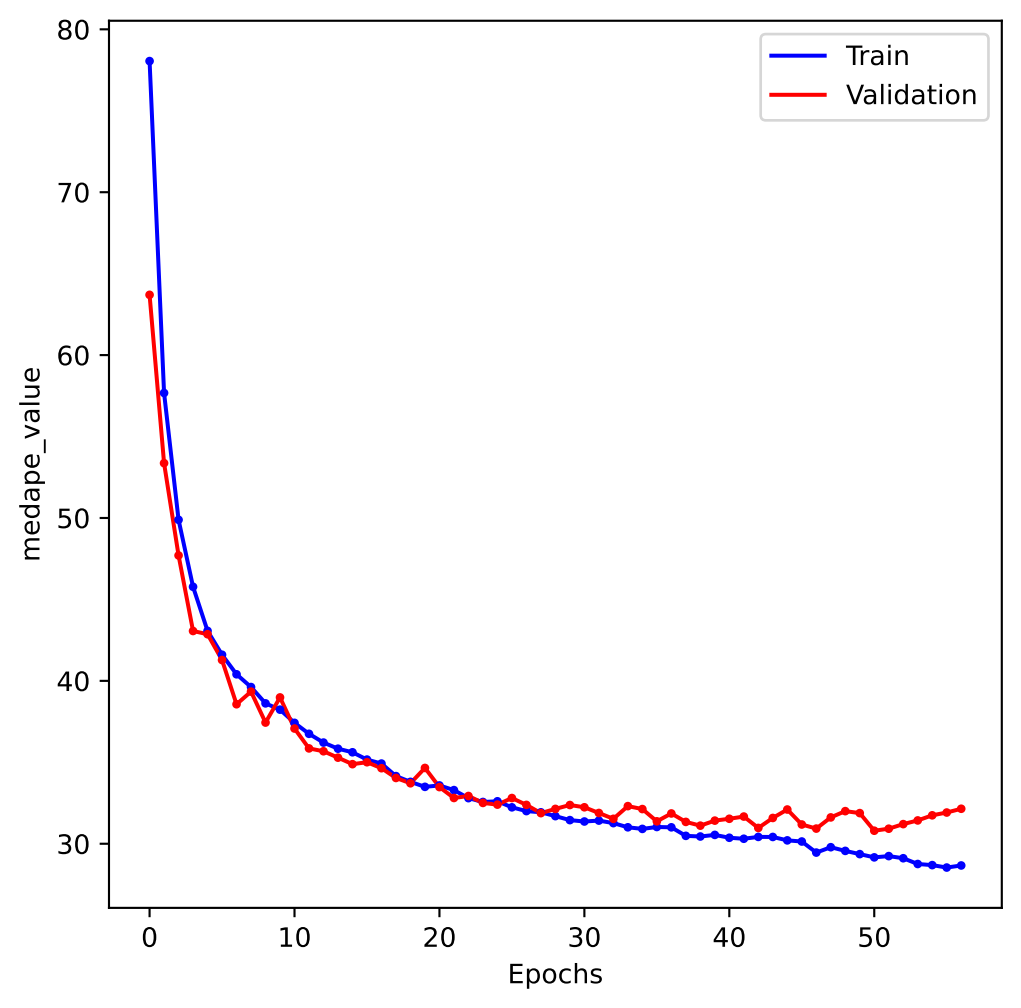

Train
Validation
loss_combined
1.2
1.1
1.0
0.9
0.8
0.7
0.6
0.5
0
10
20
30
40
50
Epochs

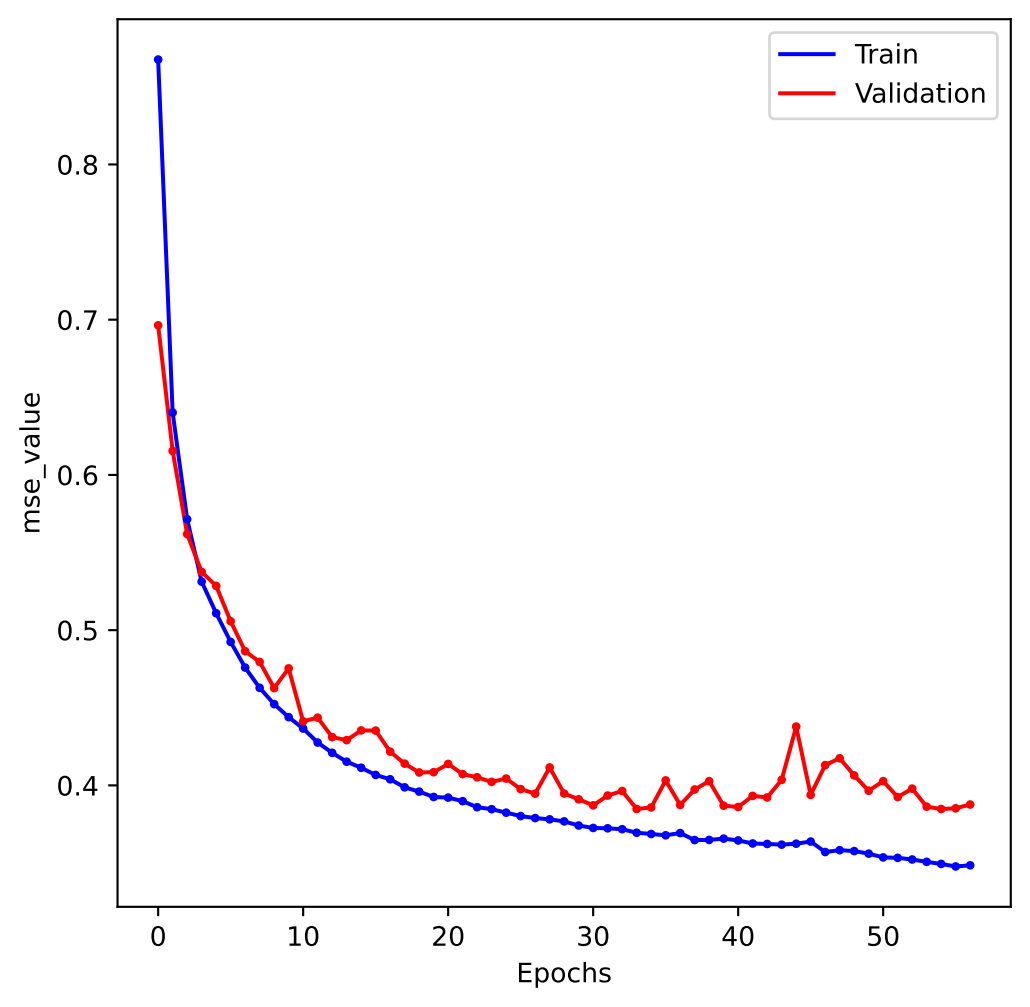

Train
Validation
loss_lower
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50
Epochs

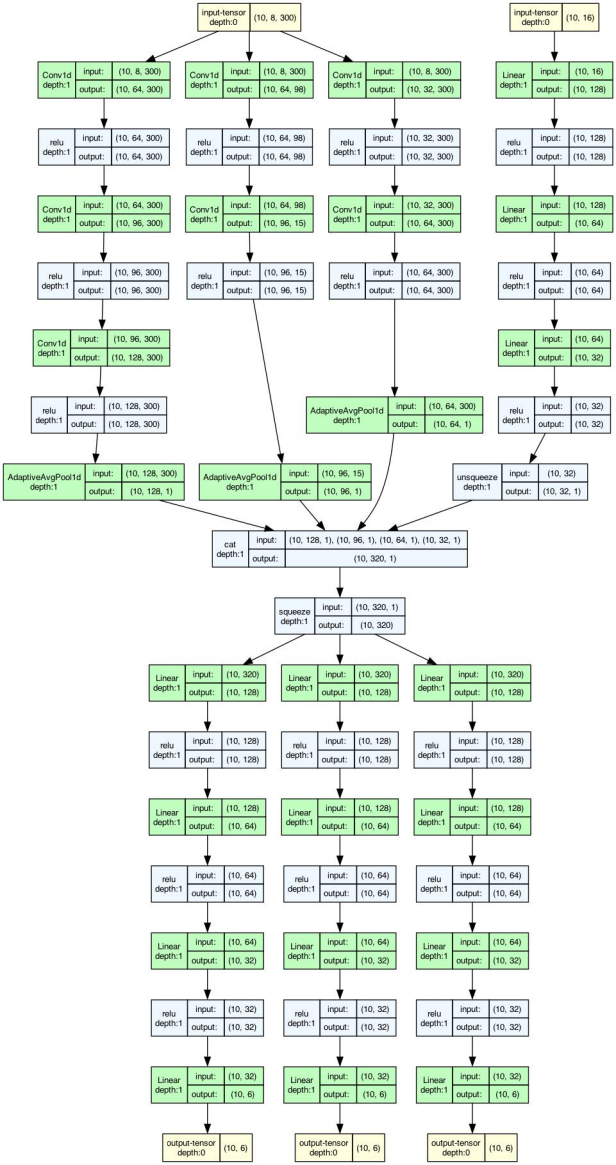

Train
Validation
0.25
0.20
0.15
0.10
loss_upper
0
10
20
30
40
50
Epochs


Train
Validation
loss_value
Epochs
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0
10
20
30
40
50


Train
Validation
mae_value
0.65
0.60
0.55
0.50
0.45
0.40
0.35
0
10
20
30
40
50
Epochs


Train
Validation
medape_value
Epochs
80
70
60
50
40
30
0
10
20
30
40
50


Train
Validation
mse_value
0.8
0.7
0.6
0.5
0.4
0
10
20
30
40
50
Epochs


input-tensor depth:0
(10, 8, 300)
input-tensor depth:1
(10, 16)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 300)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 64, 98)
Conv1d depth:1
input: (10, 8, 300)
output: (10, 32, 300)
Linear depth:1
input: (10, 16)
output: (10, 128)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64, 98)
output: (10, 64, 98)
relu depth:1
input: (10, 32, 300)
output: (10, 32, 300)
relu depth:1
input: (10, 128)
output: (10, 128)
Conv1d depth:1
input: (10, 64, 300)
output: (10, 96, 300)
Conv1d depth:1
input: (10, 64, 98)
output: (10, 96, 15)
Conv1d depth:1
input: (10, 32, 300)
output: (10, 64, 300)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 96, 300)
output: (10, 96, 300)
relu depth:1
input: (10, 96, 15)
output: (10, 96, 15)
relu depth:1
input: (10, 64, 300)
output: (10, 64, 300)
relu depth:1
input: (10, 64)
output: (10, 64)
Conv1d depth:1
input: (10, 96, 300)
output: (10, 128, 300)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 128, 300)
output: (10, 128, 300)
AdaptiveAvgPool1d depth:1
input: (10, 64, 300)
output: (10, 64, 1)
relu depth:1
input: (10, 32)
output: (10, 32)
AdaptiveAvgPool1d depth:1
input: (10, 128, 300)
output: (10, 128, 1)
AdaptiveAvgPool1d depth:1
input: (10, 96, 15)
output: (10, 96, 1)
unsqueeze depth:1
input: (10, 32)
output: (10, 32, 1)
cat depth:1
input: (10, 128, 1), (10, 96, 1), (10, 64, 1), (10, 32, 1)
output: (10, 320, 1)
squeeze depth:1
input: (10, 320, 1)
output: (10, 320)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
Linear depth:1
input: (10, 320)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
relu depth:1
input: (10, 128)
output: (10, 128)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
Linear depth:1
input: (10, 128)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
relu depth:1
input: (10, 64)
output: (10, 64)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
Linear depth:1
input: (10, 64)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
relu depth:1
input: (10, 32)
output: (10, 32)
Linear depth:1
input: (10, 32)
output: (10, 6)
Linear depth:1
input: (10, 32)
output: (10, 6)
Linear depth:1
input: (10, 32)
output: (10, 6)
output-tensor depth:0
(10, 6)
output-tensor depth:0
(10, 6)
output-tensor depth:0
(10, 6)